# Web-Based Real-Time Bus Tracking System for Enhanced Commuter Experience and Efficient Fleet Management

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***Abstract*— A web-based real-time bus tracking system is proposed to significantly improve the efficiency, reliability, and user experience of public transportation. By integrating GPS, web technologies, and real-time analytics, the system provides live updates on bus locations, enabling passengers to plan their journeys more accurately and reduce waiting times. The platform introduces predictive arrival estimations and dynamic delay notifications, enhancing the overall commuter experience and making public transit more user-friendly. For transport authorities and operators, the system delivers actionable performance insights, supporting data-driven decisions for route optimization, scheduling, and vehicle maintenance. Additionally, the application contributes to passenger safety by enabling rapid response during emergencies. Collected real-time and historical data can be analyzed using novel data visualization and trend analysis methods to improve service quality and inform long-term transport planning. This project demonstrates how smart mobility solutions leveraging real-time data and predictive analytics can address key challenges in urban transportation, promoting a more connected, responsive, and intelligent transit network.**

**Keywords— Real-Time Bus Tracking, Public Transportation, Web-Based Application, GPS Integration, Smart Mobility, Fleet Management, Intelligent Transport Systems.**

## I. INTRODUCTION

Public transportation plays a pivotal role in urban life, yet commuters often face challenges such as unpredictable traffic, varying bus schedules, and the absence of real-time information. These issues lead to inefficiencies, extended wait times, and uncertainty. To address these challenges, this project presents a Real-Time Bus Tracking Web Application designed to enhance the commuter experience by providing live updates on bus locations and arrival times. The decision to develop this application stems from the need to modernize urban transportation systems. The project leverages cutting-edge technologies, including Node.js, Express, WebSocket, and mapping libraries like Google Maps API, to deliver real-time data with high scalability and accuracy. By implementing a user-centric design, the application aims to make the transit system more accessible and efficient for both daily commuters and occasional riders. The interactive map allows users to track buses in real-time, plan their journeys, and minimize waiting times. Furthermore, the system integrates robust security measures to ensure that only authorized buses can update their locations, while user data is protected according to privacy regulations [1]. The project also explores the potential for data analytics, offering insights into traffic patterns and bus performance, which can help improve operational efficiency and decision-making for transit authorities. This Real-Time Bus Tracking Web Application is a step towards building smarter, more efficient urban transportation systems, contributing to the ongoing evolution of smart cities and improving the overall reliability and user experience of public transit.

## II. RELATED WORKS

The deployment of real-time bus tracking systems has seen rapid advancement due to the growing reliance on smart mobility solutions and IoT technologies. A considerable body of work has focused on improving commuter convenience, reducing waiting time, and enhancing fleet management through GPS integration, mobile applications, and artificial intelligence.

Balani and Mohammed (2023) [1] proposed a web-based bus tracking system built on the Internet of Things (IoT) architecture, where GPS modules interfaced with web services provide live tracking of buses through an interactive map interface. Their system offered route optimization and estimated time of arrival (ETA) features, aimed at minimizing commuter uncertainty.

GSRTC's real-time bus tracking initiative in 2024 [2] is a large-scale practical implementation that allows users to monitor the live locations of over 5,000 buses across Gujarat via a mobile-friendly platform. The system serves as a benchmark in public transportation modernization in India by improving accessibility and transparency in bus scheduling.

In an academic initiative, students at Tishk International University developed a mobile application for tracking college buses in Erbil (2024) [3]. The project focused on providing real-time location data and notifications to students and staff, demonstrating the effectiveness of localized, campus-specific transport solutions.

Sree et al. (2024) [4] introduced "BusWay," a real-time college bus tracking system that combines GPS and cloud-based infrastructure to provide dynamic location updates and estimated arrival times. The study highlighted the potential of such systems to enhance operational efficiency in educational institutions.

Elijah et al. (2023) [5] explored an innovative method for public bus fleet tracking using Bluetooth Low Energy (BLE) beacons. Unlike GPS-based systems, this approach reduces power consumption and improves indoor tracking accuracy, which is beneficial in dense urban environments with poor satellite visibility.

From an AI perspective, Rashvand et al. (2025) [6] implemented a neural network-based model to predict real-

time bus departure times using historical data and real-time transit updates. The proposed solution improved schedule reliability, which is critical in developing responsive and adaptive transportation networks.

Furthermore, the BMTC's upgraded "Namma BMTC" mobile app, launched in 2025, showcases real-world application of these technologies, enabling passengers to track over 5,200 buses, plan routes, and get fare calculations, while also offering an SOS feature for emergencies[7].

These studies collectively emphasize the ongoing trend toward integrating IoT, cloud computing, and artificial intelligence in the design and deployment of scalable real-time bus tracking systems, enhancing public transportation reliability and commuter satisfaction.

## III. OBJECTIVE

This study focuses on building a Real-Time Bus Tracking Web Application aimed at making daily commuting smoother, faster, and more reliable. The main goal is to create a smart, user-friendly platform where people can easily track buses in real time and plan their trips more efficiently.

To achieve this, the app will feature live location tracking where each bus sends its current position to a central system. An interactive map will show bus movements, and smart algorithms will estimate arrival times with accuracy. The interface will be simple and accessible, ensuring a positive experience for both regular and occasional users.Security is also a top priority, with authentication systems to protect data and ensure only verified buses access the system. The backend will use scalable technologies like Node.js, and WebSockets will enable instant updates. Optional features like notifications and data analysis will add value for users and transit authorities alike. Overall, this project aims to bring real improvement to public transportation through modern, real-time solutions.

## IV. METHODOLOGY

The application provides the live location of the bus to the user. The commuters can not only fetch the bus location but also know estimate time taken by bus to reach its destination. The Location information is fetched from the online database which receives the data regarding the location from a separate application used by drivers/conductors on the bus. This helps in maintaining the uniqueness of the bus while displaying its location on the map. The request made by the client for the bus information will be fetched from the database and delivered to client through server. The driver/conductor will send its coordinates continuously to our server where data will be stored. When the user selects that particular Bus ID, its location will be retrieved from the server and shown on the map. Since the coordinates will be changing, the point on the map will keep on moving, hence the user can actually see the live location of the selected bus. Also we will use google's distance matrix algorithm to show the user the approximate time taken by bus to reach the user. The application will be developed using **visual studio code** which has a very simple User Interface to use it. Google maps API is the core component that will be used in it, which is very easy to use and explore maps with simple gestures such as pinch to zoom tap to point etc. It will make tracking the bus very easy for the user[8].

**Proposed Plan with Prototype:** Prototype interface should be look like this when we implement the web application into android app. Our plan is to implement the whole project in mobile app version in the future. For this reason, here we are sharing the prototype concept which shown that the how our app will look like.

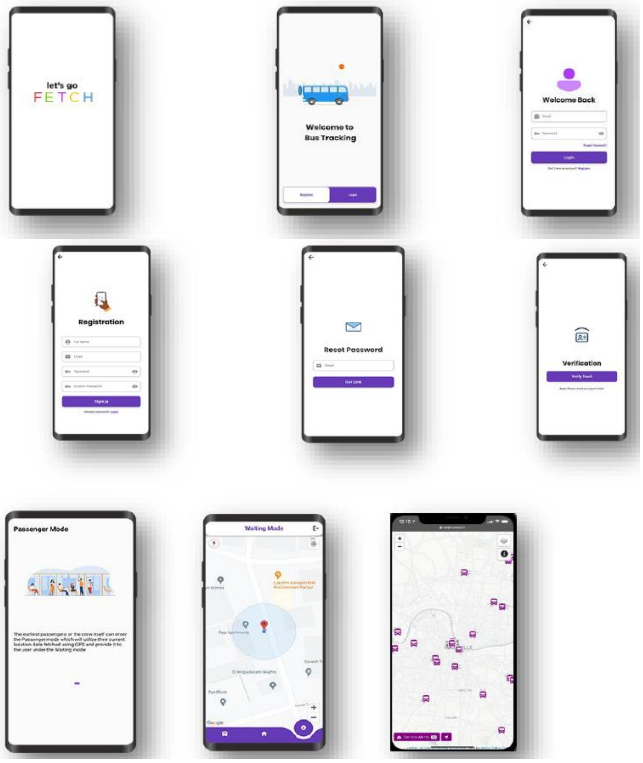



Fig. 1. Web application

**App Feature and Details:** The Bus Tracking web application offers several key features to enhance user experience. The Current Location feature requests permission to access the device's GPS, providing the user with their accurate real-time location. The Destination feature allows users to easily select their destination by typing the name, with an auto-suggestion tool to help find the right location. In the Driving Mode section, users can choose their preferred travel mode: personal vehicle, on foot, or bus. The Route feature offers three route options, displaying estimated travel time and distance to help users select the best route, including forward, backward, or both routes on the map. The Map feature includes interactive tools such as a pegman to view street-level images, satellite view, and zoom options. Users can also remove nearby place markers for a cleaner map display, making navigation and tracking easier. These features work together to provide a comprehensive and user-friendly experience[9].

**Platform and Technologies:** The Bus Tracking web application is developed using a combination of key platforms and technologies to ensure robust performance and a seamless user experience. The application is built using Visual Studio Code, providing a versatile development environment for easy access and editing. The JavaScript programming language is used to add dynamic functionality and interact with the Document Object Model (DOM) for a responsive user experience. CSS is employed for styling the

web pages, ensuring consistency and visual appeal across various devices[10]. HTML serves as the foundational structure of the web content, enabling the application to render properly in web browsers. Python is used to handle backend logic and debugging, ensuring smooth operation and efficiency of the application.

**Architecture:** The application follows a layered architecture to maintain scalability and functionality. The frontend is developed using HTML, with CSS for styling and JavaScript for implementing core functionalities. The user interface offers features like current location tracking, destination selection, route options, and real-time map views, enabling users to easily track buses and view estimated times of arrival.

**Proposed Strategy:**

Diagram: High-level system diagrams help visualize the overall architecture, illustrating how different components interact. This includes the web server, database, real-time tracking module, user interface, and external interfaces with mapping services. User interface diagrams (e.g., wireframes or mockups) are crucial for planning the layout and visual elements of the web application. They help in positioning components like the map, bus icons, and user controls. Flow diagrams help illustrate the movement of data within the system[11].

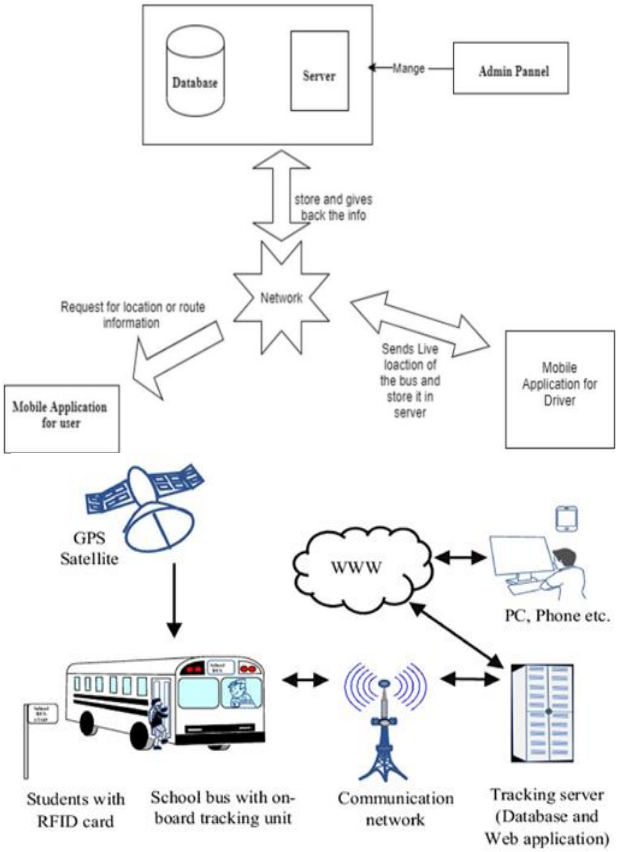



Fig. 2. Data Flow Diagram

This is crucial for understanding how real-time bus location updates are collected, processed, and communicated to end-users. Diagrams can depict the communication protocols and data formats used for interactions between the web server, buses, and the user interface. This is especially important for real-time data exchange. Diagrams can be used to visualize bus routes, stops, and potential variations based on real-time conditions. This aids in optimizing the efficiency of the bus system. Diagrams can illustrate the error-handling processes, depicting how the system reacts to unexpected events, disruptions, or errors in real-time tracking.

Algorithms: Assist in designing the logic and flow of the real-time tracking process. This involves defining how bus location data is collected, processed, and transmitted to the user interface. In the context of the user interface, algorithms may involve defining how the map interacts with real-time data, handling user interactions, and updating the display based on the received information. Algorithms detail the steps involved in processing real-time data, such as determining bus positions, calculating estimated arrival times, and updating the user interface in response to new information. Algorithms define the methods and protocols for communication, specifying how data is transmitted securely between different components of the system. Algorithms may involve route optimization based on live traffic data, allowing buses to dynamically adjust their routes to minimize delays and improve overall efficiency. Algorithms play a role in implementing security features, defining how user authentication is handled, ensuring secure data transmission, and protecting against potential vulnerabilities. Algorithms define the logic for detecting errors, handling exceptions, and recovering from failures, ensuring the system remains robust and reliable[12].

**Duration Calculation Algorithm:**

Duration is commonly calculated as the difference between two time points. The specific algorithm depends on the format of your time representation and the granularity of the duration you need (e.g., seconds, minutes, hours).

**Step 1:-** Obtain the starting time as (h1, m1, s1) and the ending time as (h2, m2, s2)

**Step 2:-** Use mathematics Calculation startSeconds = h1 * 3600 + m1 * 60 + s1 endSeconds = h2 * 3600 + m2 * 60 + s2 durationSeconds = endSeconds - startSeconds

**Step 3:-** Convert duration to hours, minutes, and seconds durationHours = floor(durationSeconds / 3600) durationMinutes = floor((durationSeconds % 3600) / 60) durationSeconds %= 60 return (durationHours, durationMinutes, durationSeconds)

## V. IMPLEMENTATION RESULTS AND ANALYSIS

In this application is designed to provide users with up-to-date information about the location and status of buses in a public transportation system.

**Live Bus Location:** Users can view the real-time location of buses on a map. The application continuously updates the bus positions, providing accurate and timely information.

**Estimated Arrival Times:** Users can access estimated arrival times for buses at specific stops. Predictions are based on the current bus locations and historical data, helping users plan their journey more efficiently.

**Route Information:** Detailed information about bus routes. Users can explore the entire route map and understand where each bus is headed.

**User-Friendly Interface:** The application has an intuitive and user-friendly interface, making it easy for users to navigate and access relevant information[13].

### A. *Uniqueness of Bus Tracking Web Application*

This web application lies in its ability to provide a dynamic, up-to-the-minute view of the public transportation system, offering several distinctive features and advantages:

**Live Updates and Accuracy:** This application provides real-time bus location updates, ensuring users have accurate and current information about bus positions and arrival times.

**Enhanced User Experience:** Users can plan their journeys more efficiently by knowing exactly where buses are and when they will arrive. This contributes to a smoother and more predictable commuting experience.

**Multi-Model Integration:** The application may integrate with multiple modes of transportation, providing a holistic view of the transit network. This could include information about buses, trains, metros, and other services.

**Predictive Analytics:** Utilizing historical data and predictive analytics, the application can estimate arrival times more accurately, even during periods of high traffic or unusual events.

**Privacy and Security Measures:** Implementing robust privacy and security measures, such as anonymizing user data and securing communication channels, distinguishes the project as being attentive to user privacy.

**Scalability and Reliability:** The application is designed to scale seamlessly with an increasing number of users and buses. The reliability of real-time updates even during periods of high demand sets it apart.

**Integration with Smart City Initiatives:** Integration with broader smart city initiatives, such as intelligent traffic management or environmental monitoring, positions the project as part of a larger effort to improve urban living.

**Open Data Standards:** Supporting open data standards and APIs encourages collaboration and interoperability with other services and applications, fostering innovation and a more connected urban ecosystem[14].

### B. *Comparison of this project with Uber*

This web application and Uber both fall under the umbrella of transportation services, they serve different purposes and have distinct characteristics. Let's compare these two and see:

**Public Transportation Focus:** Primarily focuses on providing real-time information about public transportation systems, such as buses. Targets a broader audience, including regular commuters, tourists, and individuals who rely on public transit.

**Fixed Routes and Schedules:** Operates on fixed routes and schedules defined by the public transportation authority. Provides users with information about bus routes, stops, and schedules.

**Government or Municipal Involvement:** Typically operated or supported by government or municipal transportation authorities. Integrates with city planning and public transportation management.

**Affordability:** Generally more affordable for users compared to individual ride-hailing services. Focused on making public transportation accessible to a wide range of people.

**Community Impact:** Contributes to reducing traffic congestion, environmental impact, and overall community mobility. Supports city planning initiatives and is aligned with public transportation goals.

On the other side,

**On-Demand Ride-Hailing:** Primarily provides on-demand ride-hailing services, allowing users to request private rides. Targets individuals who prefer door-to-door, on-demand transportation.

**Dynamic Pricing:** Utilizes dynamic pricing based on demand and supply. Users may experience price fluctuations during peak hours or high-demand periods.

**Flexible Routes and Personalized Service:** Offers flexibility in routes and destinations, catering to individual user preferences. Provides a personalized experience where users can choose specific drop-off locations.

**Private Enterprise:** Operated by a private company as a for-profit service. Operates independently of municipal transportation authorities.

**Global Reach:** Operates in numerous cities globally, providing a consistent service experience across different locations[15].

**Technology and Innovation:** Utilizes advanced technology for ride matching, navigation, and seamless payment options. Often introduces new features and services, such as Uber Eats for food delivery[16].

### C. *UI/UX Design and User Interaction*

The UI/UX design offers an intuitive interface with real-time bus tracking on an interactive map, allowing users to easily select destinations and view multiple route options. The design focuses on accessibility, responsiveness, and user-friendly controls for a seamless experience.

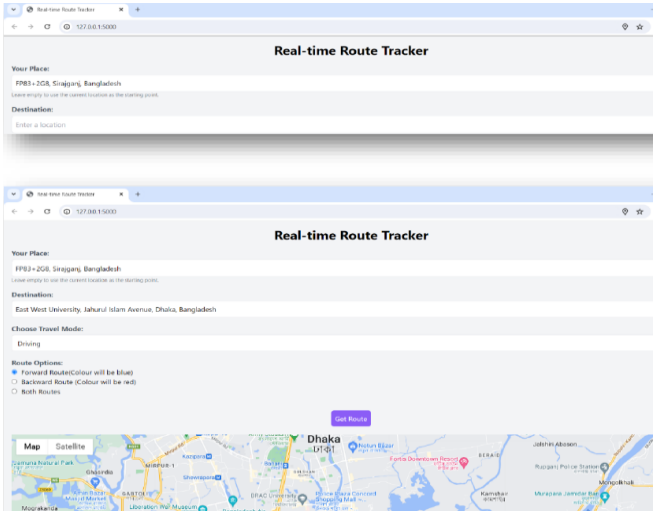



Fig.3. Real-time route tracker

The application uses Google's Distance Matrix API to calculate the distance and estimated arrival time between the user's location and the bus's destination. It updates in real-time, providing accurate predictions as the bus moves.

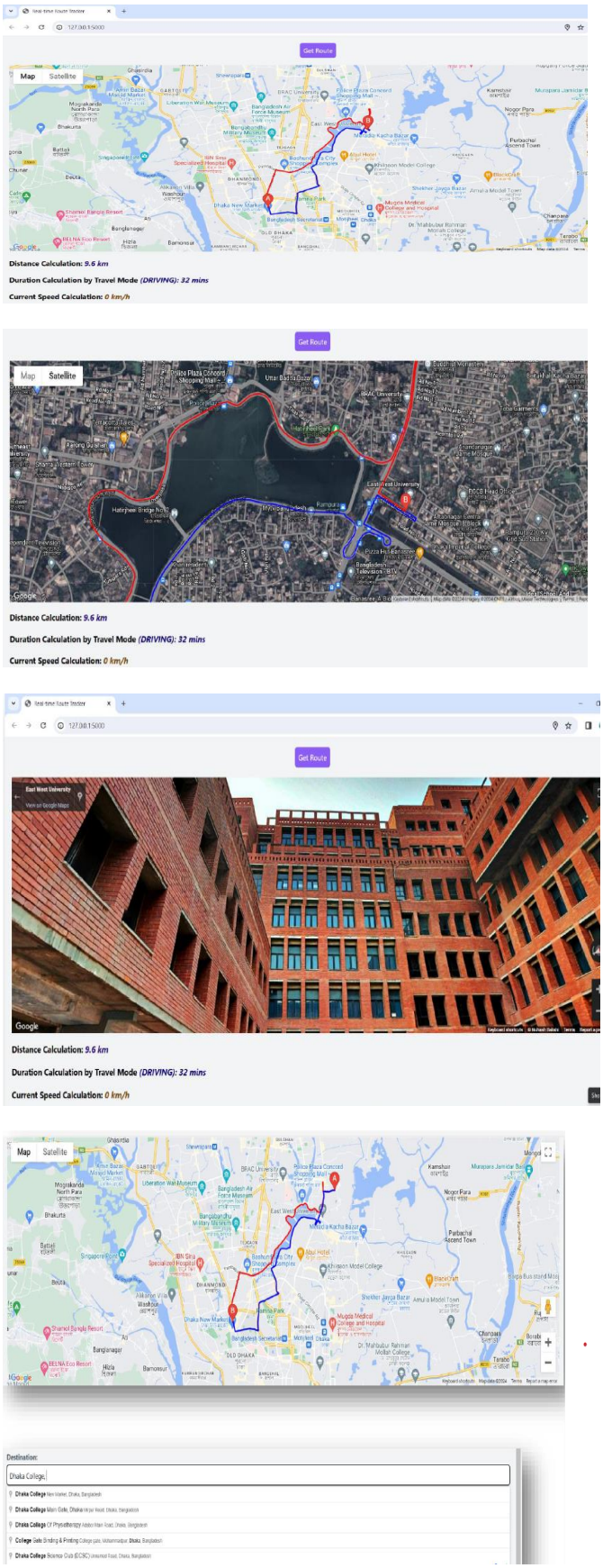



Fig. 4. Distance Calculation

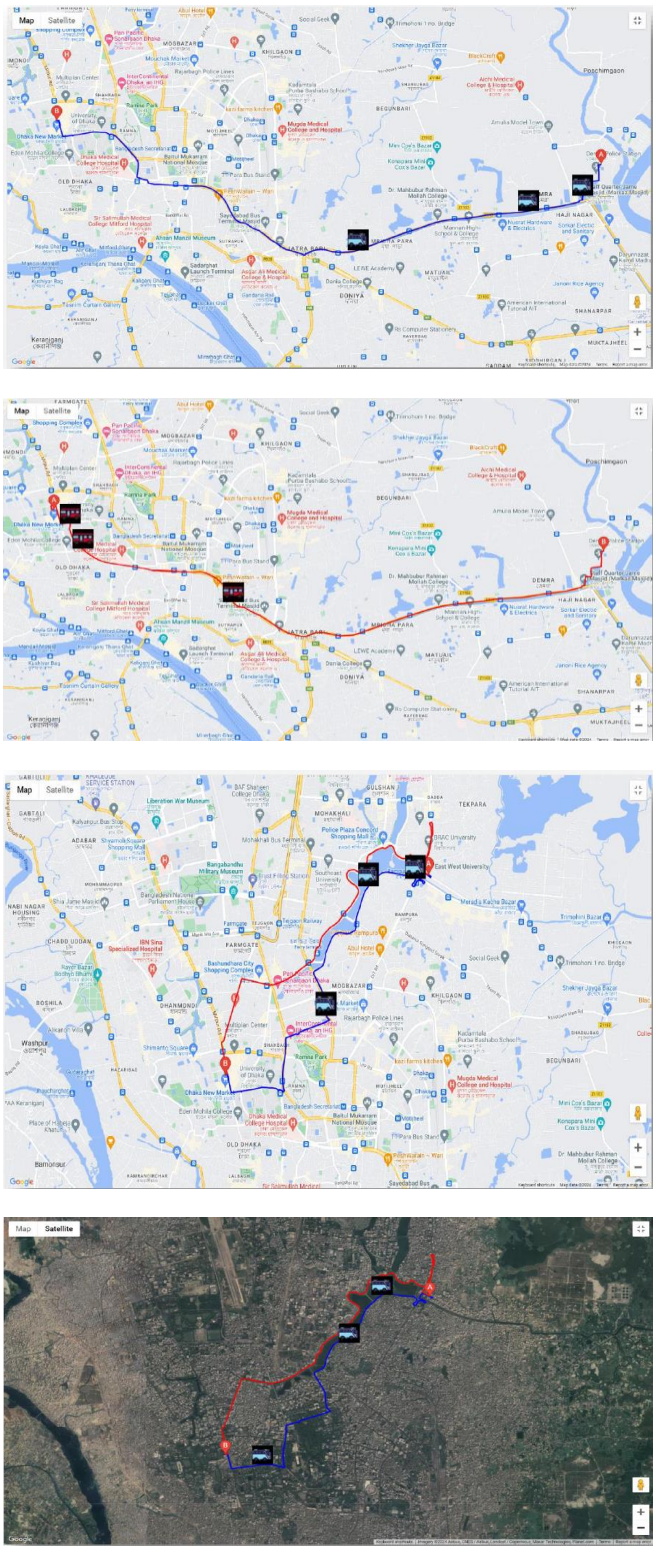



Fig. 5. Real-Time Bus Tracking

The Bus Tracking application leverages real-time tracking on an interactive map to provide users with accurate and dynamic bus location updates. By using satellite view, the system offers a detailed and realistic representation of the bus's movement along its route. The bus's position is displayed with live markers that update as the bus progresses, offering users a precise view of the bus's location in real-time. The satellite view enhances this experience by displaying the surrounding environment, giving users

contextual information about the route. The map interface is highly interactive, allowing users to zoom in, zoom out, and navigate the route for a better understanding of the bus's path. This feature ensures that commuters have up-to-date information, improving convenience, efficiency, and overall satisfaction with the system. The integration of real-time tracking with satellite imagery optimizes the bus tracking process, making it more reliable and user-friendly.

## VI. Conclusion and Future Work

This Real-Time Bus Tracking Web Application presents an innovative approach to improving public transportation by offering a user-friendly platform that delivers live updates, accurate arrival estimates, and route information. By integrating technologies such as GPS, real-time communication, and predictive analytics, the system empowers commuters to make informed decisions, reducing uncertainty and waiting time. This contributes to a more reliable, efficient, and accessible transit experience, supporting broader goals of smart city development and sustainable urban mobility. Despite its benefits, the project faces some limitations. Real-time tracking accuracy depends on GPS signal quality, which can be disrupted in dense urban areas. Delays in data transmission may cause lags in live updates. Additionally, the system requires stable internet access and compatible devices, which may limit accessibility for some users. Scalability challenges, privacy concerns, and the need for continuous maintenance are also important factors to address. Looking ahead, future development will focus on expanding integration with other transit modes, including trains and shared mobility services, to offer a more connected travel experience. Collaboration with smart city infrastructure can help optimize routing based on live traffic data. Planned features include multilingual support, offline access to schedules, user-submitted feedback, enhanced accessibility, and environmental impact tracking. Integrating digital fare payments and gamification elements may further improve user engagement and adoption. The ultimate goal is to develop a full-featured Android application, making the platform even more accessible and convenient for a wider audience. By continuously improving and adapting to emerging technologies and user needs, this project aims to drive innovation in public transportation and contribute meaningfully to the creation of smarter, more inclusive cities.

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